Lecture 12

Orientation, integration and Stokes' theorem

Objectives. *An orientation specifies the sign of a top-degree form. It lets us integrate forms on a manifold. Stokes' theorem identifies the integral of an exterior derivative with the integral on the boundary, using outward-first orientation. We prove it in every dimension and apply it to the flux of the Coulomb form.*

By the end of this lecture you will be able to:

- Determine orientability using charts and top forms.
- Define an integral independent of the oriented charts and partition of unity.
- Prove Stokes' theorem with its support and orientation hypotheses.
- Compute Coulomb flux by removing a ball around its singularity.

12.1 Orientations and oriented charts

For an n-dimensional real vector space with $n \geq 1$, two ordered bases have the same *orientation* if their change-of-basis determinant is positive. There are two classes. A *frame* on an open set is an ordered list of smooth tangent fields that is a basis at every point. An *orientation of a manifold* chooses a class in each tangent space, locally represented by a smooth frame. A manifold is *orientable* if it admits such a choice, and *oriented* when a choice has been made.

An *oriented atlas* has positive Jacobian determinants on all chart transitions. Its coordinate frames give compatible orientation classes: the change of frame uses the inverse Jacobian, whose determinant is also positive. Conversely, near any point an orientation is represented by a frame. Its determinant relative to a coordinate frame is continuous and nonzero. Shrink to a connected chart so that the sign is constant. If necessary replace x^1 by $-x^1$. The coordinate frame then represents the chosen orientation. Charts obtained this way have positive transitions. This proves the equivalence of orientations and oriented atlases.

A nowhere-zero top form μ selects the bases on which its value is positive. These choices are locally represented by coordinate frames, because its coefficient in a connected chart has constant sign. For oriented manifolds, a diffeomorphism is *orientation-preserving* when its differential sends positive bases to positive bases. In positive charts this means its Jacobian determinant is positive.

For example, on $S^n \subset \mathbb{R}^{n+1}$ the form

$$\mu_p(v_1, \dots, v_n) = \det[p\ v_1\ \cdots\ v_n] \qquad (v_j \in T_pS^n)$$

is smooth and nowhere zero: the radial vector p is independent of the n tangent basis vectors. It defines the outward orientation. For $n = 2$ it is the sphere area form of Lecture 10.

Lecture 10 computed the north-pole stereographic inverse pullback as $-4\mathrm{d}u\wedge\mathrm{d}v/(1+u^2+v^2)^2$. This chart reverses the outward orientation.

In dimension zero an orientation is a choice of sign $+1$ or -1 at each point. Integration of a function then means its signed sum. This convention includes the endpoints of an oriented interval.

12.2 Partitions of unity and orientability

A collection of sets is *locally finite* if every point has a neighbourhood meeting only finitely many of them. A smooth *partition of unity subordinate to an open cover* $\{U_i\}$ consists of nonnegative smooth functions ρ_i such that

$$\sum_i \rho_i = 1, \qquad \operatorname{supp}\rho_i \subset U_i, \qquad \{\operatorname{supp}\rho_i\} \text{ is locally finite.}$$

One may use several functions assigned to the same cover member. Local finiteness makes each sum finite near any point. It also implies that a compact set meets only finitely many supports: choose finitely many such neighbourhoods covering the compact set.

Theorem 12.1 (Partitions of unity · *sketched; normalization and use proved*)**.** *Every open cover of a smooth manifold admits a subordinate smooth partition of unity.*

Sketch of existence. Second countability and local compactness give a countable exhaustion by compact sets, each lying in the interior of the next. Refining the cover on the successive compact layers gives a locally finite cover by precompact coordinate balls with smaller balls still covering the manifold. Choose a smooth bump on each ball that is positive on its smaller ball and supported in the larger one. The coordinate construction uses the smooth function $e^{-1/t}$ for $t>0$, extended by zero for $t\le 0$. The locally finite sum $b=\sum_i b_i$ of these bumps is smooth and strictly positive. Thus $\rho_i=b_i/b$ is smooth and nonnegative, has the required support, and sums to one. The refinement and bump construction are the sketched steps. See Tu, Theorem 13.7 and Appendix C. ∎

Proposition 12.2 (Orientability criteria · *proved given partitions of unity*)**.** *A smooth manifold is orientable if and only if it has an oriented atlas, and if and only if it admits a nowhere-zero smooth top form.*

Proof. The atlas equivalence and the top-form-to-orientation implication were proved above. For the converse, take positively oriented charts and a subordinate partition. Extend $\rho_i\mathrm{d}x_i^1\wedge\cdots\wedge\mathrm{d}x_i^n$ by zero outside its chart. This is smooth because its support lies inside the chart. Their locally finite sum is smooth. Every summand is nonnegative on a positive tangent basis, and at least one has positive coefficient at every point. The sum therefore never vanishes. In dimension zero the constant function 1 is already a nowhere-zero 0-form, and every discrete manifold has an atlas. ∎

The strip model in the figure shows the transverse direction on a Möbius band.



Figure 12.1: The red edge arrows mark $(t, s) \mapsto (t + 1, -s)$. An upward transverse arrow at $t = 1$ therefore returns at $t = 0$ pointing down.

The open Möbius band is the quotient of $\mathbb{R} \times (-1, 1)$ by this identification. On a rectangle of width less than 1, the projection is injective and open: the inverse image of its image is the open union of all translated copies under the identification. The inverse restrictions give charts on the quotient. Their transitions are restrictions of integer powers of $(t, s) \mapsto (t+1, -s)$, hence smooth. Thus the projection is a local diffeomorphism. If the quotient had a nowhere-zero top form, its pullback would be $h(t, s)\mathrm{d}t \wedge \mathrm{d}s$ with h continuous, nonzero and of constant sign on the connected strip. Invariance under the identification gives $h(t + 1, -s) = -h(t, s)$, contradicting that constant sign. Hence it is not orientable.

For $\mathbb{R}\mathrm{P}^2$, the projection $\pi\colon S^2 \to \mathbb{R}\mathrm{P}^2$ is a local diffeomorphism by the ratio charts of Lecture 7. There it was called q; we write π here. On a hemisphere $\sigma X^i > 0$, with $\sigma \in \{+1, -1\}$, its coordinate map is $X \mapsto (X^j/X^i)_{j \neq i}$. Its smooth inverse is $y \mapsto \sigma Y/\|Y\|$, where Y has ith component 1 and remaining components y. This proves the local diffeomorphism assertion. A nonzero top form would pull back to $h\mu$ on S^2, where μ is its outward area form and h has constant nonzero sign. The antipodal map $a(p) = -p$ satisfies

$$(a^*\mu)_p(v, w) = \det[-p\ -v\ -w] = -\mu_p(v, w).$$

Since $\pi \circ a = \pi$, invariance forces $h(-p) = -h(p)$, again impossible. Thus $\mathbb{R}\mathrm{P}^2$ is not orientable.

12.3 Integrating a top form

Recall the ordinary change-of-variables theorem: under a smooth diffeomorphism $y = f(x)$, integration of a compactly supported function uses the factor $|\det Df|$. In positive coordinates the determinant itself is positive. This is the calculus input to integration of forms.

For $n \geq 1$, let ω be a smooth compactly supported n-form on an oriented n-manifold. On a positive chart $x\colon U \to x(U)$ write $(x^{-1})^*\omega = f\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n$. If its support lies in U, define its integral to be the ordinary integral of f on $x(U)$. For a general support, choose a positive atlas and a subordinate partition and set

$$\int_M \omega = \sum_i \int_{U_i} \rho_i \omega.$$

Only finitely many terms meet the compact support. Each has compact support inside its chart, so every term is defined.

Proposition 12.3 (Independence of choices · *proved given partitions of unity and change of variables*)**.** *The integral is independent of the positive atlas and subordinate partition. Reversing the orientation reverses its sign.*

Proof. On overlapping positive charts, the determinant rule for top-form pullback and ordinary change of variables give the same chart integral. For two partitions ρ_i and χ_j subordinate to positive charts, insert $\sum_j \chi_j = 1$ into each integral for the first partition. The result is the finite double sum of the integrals of $\rho_i \chi_j \omega$. Each term is supported in the chart overlap and has the same value in either chart system. Interchanging the sums and using $\sum_i \rho_i = 1$ gives the second partition's answer. This proves independence. For the opposite orientation, reflect one coordinate. Its derivative has determinant -1, whereas the absolute Jacobian in ordinary integration is 1. The top-form coefficient changes sign, proving the last assertion. ∎

For computations, let $D \subset \mathbb{R}^n$ be a compact parameter region, and let $F\colon D \to M$ be smooth with $F(D) = M$. Here smoothness on D means local smooth extension to open neighbourhoods in $\mathbb{R}^n$. Suppose F restricts to an orientation-preserving diffeomorphism $\operatorname{int} D \to M \setminus E$, where $E = F(\partial D)$. Assume ∂D has Lebesgue measure zero and E has measure zero in every manifold chart. The latter condition is chart-independent by change of variables. Then integration is computed by pullback:

$$\int_M \omega = \int_D F^* \omega.$$

Indeed, insert a subordinate partition ρ_i on M. For each summand, ordinary change of variables on $\operatorname{int} D$ gives the integral of $\rho_i \omega$ on $M \setminus E$. The two null boundaries add zero. Summing the finitely many nonzero chart integrals proves the formula. These are the hypotheses behind the angular rectangle used below; see also Tu, §23.4, Definition 23.10.

A *density* has local expressions $f|\mathrm{d}x^1 \cdots \mathrm{d}x^n|$ that transform with the absolute Jacobian. Densities can be integrated without orientation because the ordinary change-of-variables factor has the same law. A top form uses a signed determinant, so its signed integral requires a chosen orientation. The two transformation laws must be distinguished on nonorientable manifolds.

12.4 Manifolds with boundary

The upper and lower closed half-spaces are

$$\mathbb{H}^n = \{x \in \mathbb{R}^n : x^n \geq 0\}, \qquad \mathbb{H}^n_- = \{x \in \mathbb{R}^n : x^n \leq 0\}.$$

A smooth *manifold with boundary* is Hausdorff and second countable, with charts onto relatively open subsets of either half-space. Allowing either side permits the coordinate reflection needed for positive charts also in dimension one. For example, at the right endpoint of the oriented interval $[0, 1]$, the positive chart $u = x - 1$ maps into $(-\infty, 0]$. Chart transitions and their inverses extend smoothly to neighbourhoods in $\mathbb{R}^n$ near each point. The *boundary* ∂M consists of chart points with $x^n = 0$; the others form the interior. This geometric boundary differs from Lecture 6's boundary operator on chains. The set of boundary points is independent of the chart. Indeed, the chain rule on interior points and continuity make transition derivatives invertible also at boundary points. The inverse function theorem then makes their extensions local diffeomorphisms. Such a map cannot send a half-neighbourhood to a full interior neighbourhood: its inverse would have to take points on both sides of the source hyperplane into the half-space. Restricting charts to $x^n = 0$ gives charts of dimension $n - 1$ on ∂M. The restricted transitions make them smoothly compatible. A transition's last component vanishes on the boundary, so its tangential derivatives vanish there. Its

normal derivative is nonzero by invertibility. It sends inward vectors to inward vectors because it sends interior points to interior points. Thus the outward transverse direction is chart-independent, even when the two charts use opposite half-spaces.

At a boundary point, T_pM is the ambient n-dimensional space modeled by a half-space chart, with changes given by the transition Jacobians. Its vectors need not be velocities of two-sided curves remaining in M. In particular the outward vector $-\partial_n$ is allowed in the upper model. Functions and forms are smooth when their chart coefficients extend smoothly to open sets of $\mathbb{R}^n$. Their derivatives at the boundary are independent of extension, since extensions agree on the interior and all their derivatives agree there and extend continuously to the boundary. The same half-chart construction gives tangent changes and exterior differentiation. Partitions of unity and integration extend using restrictions of bumps to half-balls. The preceding normalization and independence proofs then apply.

On an oriented manifold the *induced boundary orientation* is *outward first*: a boundary basis $(v_1, \dots, v_{n-1})$ is positive if $(\nu, v_1, \dots, v_{n-1})$ is positive in T_pM, where ν is outward transverse. Changing ν by a tangent vector or multiplying it by a positive scalar does not change this sign. This proves independence of that choice; no metric is required. For the standard orientation on $\mathbb{H}^n$, take $\nu = -\partial_n$. The boundary orientation is represented by

$$\iota_{-\partial_n}(\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n) = (-1)^n \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1} \quad \text{on } x^n = 0.$$

On $\mathbb{H}^n_-$ the outward vector is $+\partial_n$, so the boundary orientation is $(-1)^{n-1}\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1}$. For $n = 1$ the upper model gives the negative initial endpoint, and the lower model gives the positive final endpoint. The figure compares the upper half-plane with an interval.

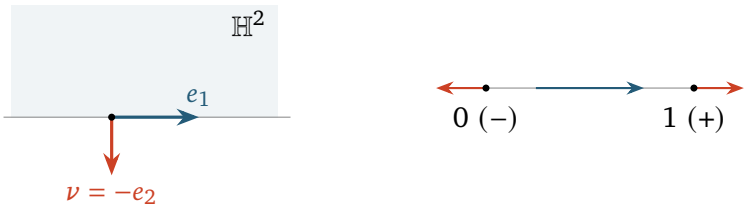


Figure 12.2: On the upper half-plane, the outward vector points down. The boundary tangent points right. On the positively oriented interval, the induced endpoint signs are − and +.

The sphere orientation above is exactly the boundary orientation of the unit ball. The usual Möbius band with its boundary is also nonorientable, since an orientation would restrict to its nonorientable interior. A *closed manifold* means a compact manifold without boundary.

12.5 Stokes' theorem in every dimension

Theorem 12.4 (Stokes · *proved given partitions of unity*)**.** *Let M be an oriented smooth n-manifold with boundary, where $n \geq 1$. For a smooth compactly supported $(n-1)$-form ω,*

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega.$$

The right side means restriction by the boundary inclusion, with the outward-first orientation. In particular the statement applies to every smooth $(n-1)$-form when M is compact.

Proof. First work on $\mathbb{H}^n$ with compact support. Write the form as

$$\omega = \sum_{i=1}^{n} (-1)^{i-1} f_i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n, \qquad \mathrm{d}\omega = \Big(\sum_i \partial_i f_i\Big) \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

The derivative formula follows because repeated coordinate factors vanish; the two omission signs cancel for the surviving derivative. Fubini's theorem and the fundamental theorem of calculus integrate $\partial_i f_i$ first in x^i. For $i < n$ the endpoints are outside compact support, so each contribution is zero. The normal contribution is

$$\int_{\mathbb{H}^n} \mathrm{d}\omega = \int_{\mathbb{R}^{n-1}} \int_0^{\infty} \partial_n f_n(x', x^n)\, \mathrm{d}x^n\, \mathrm{d}x' = -\int_{\mathbb{R}^{n-1}} f_n(x', 0)\, \mathrm{d}x'.$$

On the boundary only the term $i = n$ restricts nontrivially. Its coefficient is $(-1)^{n-1} f_n$. The boundary orientation has the additional sign $(-1)^n$, so its integral is also $-\int f_n(x', 0)\mathrm{d}x'$. For $n = 1$, the same equality uses the negative boundary point and the ordinary endpoint formula. This proves the half-space case in every dimension. On $\mathbb{R}^n$ all coordinate integrations have two endpoints outside support, so the integral of $\mathrm{d}\omega$ is zero, matching its empty boundary. On $\mathbb{H}^n_-$, integrating the normal coordinate from $-\infty$ to 0 gives $+\int_{\mathbb{R}^{n-1}} f_n(x', 0)\mathrm{d}x'$. Its boundary orientation has sign $(-1)^{n-1}$. Multiplying by the same sign in the restricted form gives $+f_n$, so Stokes holds there too. For $n = 1$ this gives the positive final endpoint.

Now choose positively oriented interior and boundary charts on M and a subordinate partition ρ_i. Each $\rho_i \omega$ has compact support inside its chart. In coordinates it extends by zero across the artificial chart edges to a smooth compactly supported form on $\mathbb{R}^n$, $\mathbb{H}^n$ or $\mathbb{H}^n_-$. The local result therefore applies to it. Sum these equalities. Local finiteness and compact support make the sum finite. On M, the product rule gives

$$\sum_i \mathrm{d}(\rho_i \omega) = \Big(\sum_i \mathrm{d}\rho_i\Big) \wedge \omega + \Big(\sum_i \rho_i\Big) \mathrm{d}\omega = \mathrm{d}\omega.$$

Here $\sum_i \rho_i = 1$ and its differential is zero. The restricted partition also sums to one on the boundary. Thus the summed local equalities give the claimed global equality. ∎

Tu, Theorem 23.12, pp. 269–271, glues in every dimension but computes locally only on $\mathbb{H}^2$. His Problem 23.4 treats $\mathbb{R}^n$ and $\mathbb{H}^n$; their proofs are supplied above.

The support hypothesis matters on a noncompact manifold. On $\mathbb{R}$, the improper integral of $\mathrm{d}(\tanh x)$ is $1 - (-1) = 2$, while the boundary is empty. The function has no compact support. The nonzero limiting values at infinity remain in the endpoint formula.

12.6 Divergence and equal fluxes

Here a *compact smooth region* in $\mathbb{R}^n$ means a compact n-manifold with boundary embedded in $\mathbb{R}^n$. For an oriented manifold S, write $-S$ for the opposite orientation.

Corollary 12.5 (Consequences of Stokes · *proved given partitions of unity*)**.** *For a smooth vector field v on a compact smooth region $K \subset \mathbb{R}^3$, with Euclidean volume $\mu = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$,*

$$\int_K (\operatorname{div} v)\mu = \int_{\partial K} \iota_v \mu.$$

On a closed oriented manifold the integral of an exact top form is zero. If an oriented compact 3-manifold has boundary $S_1 \sqcup (-S_2)$ and a closed 2-form β is smooth on it, then $\int_{S_1} \beta = \int_{S_2} \beta$.

Proof. The first statement follows by applying Stokes to $\iota_v\mu$: Lecture 11 gives $\mathrm{d}\iota_v\mu = (\operatorname{div} v)\mu$. With an outward unit normal ν and a positively oriented orthonormal boundary basis, $\mu(v,\cdot,\cdot)$ is $(v\cdot\nu)$ times $\iota_\nu\mu$, the induced boundary area form. Thus this is the ordinary normal-flux divergence theorem. The empty boundary gives the second statement. Closedness gives $\int \mathrm{d}\beta = 0$; the two induced boundary orientations give $\int_{S_1}\beta - \int_{S_2}\beta = 0$, proving the third. ∎

12.7 Coulomb flux around a missing point

The Coulomb flux form uses Lecture 10's ambient β divided by r^3; these forms agree on the unit sphere. On $\mathbb{R}^3 \setminus \{0\}$ set $r = \sqrt{x^2+y^2+z^2}$ and define the Coulomb flux form by

$$\beta = \frac{x\,\mathrm{d}y\wedge\mathrm{d}z + y\,\mathrm{d}z\wedge\mathrm{d}x + z\,\mathrm{d}x\wedge\mathrm{d}y}{r^3}.$$

Let K be a compact smooth region with outward-oriented boundary $S = \partial K$ and $0 \notin S$. We compute the outward flux of β through S. When the origin lies inside K, a removed small ball has surface S_ε. Its induced inner normal points into the ball, as the section of the region in the figure shows.

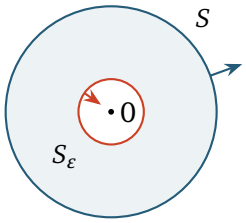


Figure 12.3: The region between S and the small sphere has an inner boundary with normal pointing into the removed ball. Its induced inner orientation is opposite to the ball's outward orientation.

Example 12.6 (Outward Coulomb flux)**.** The derivative of β has volume coefficient

$$\partial_x\frac{x}{r^3} + \partial_y\frac{y}{r^3} + \partial_z\frac{z}{r^3} = \frac{3}{r^3} - \frac{3(x^2+y^2+z^2)}{r^5} = 0.$$

Thus β is closed on its domain. On the unit sphere it is the area form of Lecture 10, so angular coordinates give

$$\int_{S^2}\beta = \int_0^{2\pi}\int_0^{\pi}\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi = 4\pi.$$

Here θ is the polar angle and ϕ the azimuth. The coordinate seam and poles have measure zero and do not change this integral. On the sphere of radius $\varepsilon > 0$, radial scaling contributes ε^3 in the numerator and ε^3 in the denominator, so its outward flux is again 4π.

If $0 \notin K$, Stokes applies directly to the closed form on K and gives zero flux. If 0 is inside K, remove an open ball of radius ε whose closure lies in the interior of K. The remaining compact region has boundary $S \sqcup (-S_\varepsilon)$. The form is smooth and closed on this region. The equal-flux consequence gives $\int_S\beta = \int_{S_\varepsilon}\beta = 4\pi$. Thus the outward Coulomb flux detects whether the original region contains the origin.

Self-check. Compute the area of the unit sphere using its angular area form.

Answer. The integral is $\int_0^{2\pi}\int_0^{\pi}\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi = 2\pi[-\cos\theta]_0^{\pi} = 4\pi$.

Summary. Orientations fix signs of top forms. Partitions of unity define a chart-independent integral. Stokes applies to compactly supported forms on oriented manifolds with boundary, using outward-first boundary orientation. Its local proof is the fundamental theorem of calculus in the normal coordinate. Closed forms have equal fluxes across two oriented surfaces that together, with the second orientation reversed, bound a compact region. Removing a ball around the Coulomb singularity gives flux 4π. Lecture 13 studies closed forms modulo exact forms.

Reading. Tu, *An Introduction to Manifolds*, §13 and Appendix C; §§21–23, especially Theorem 23.12, pp. 269–271.